

Sequential Peg-IFN after bepirovirsen may reduce post-treatment relapse in chronic hepatitis B

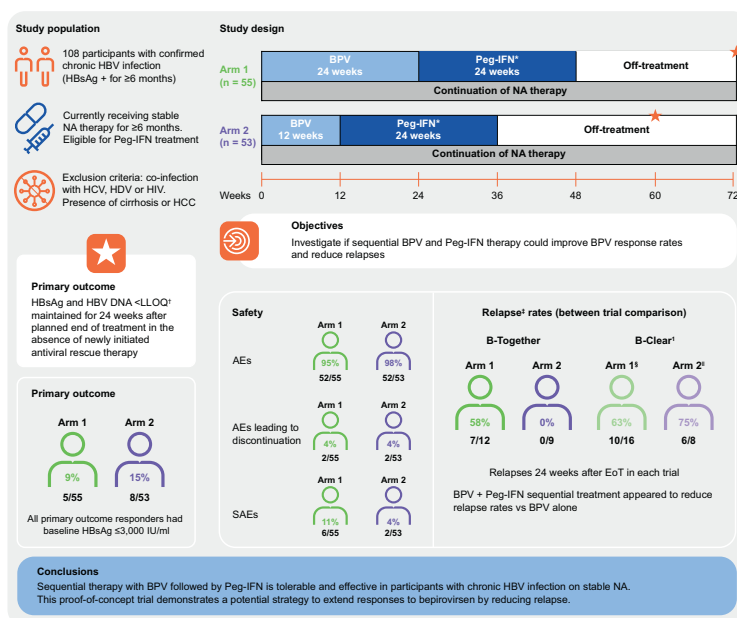
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Graphical abstract



Highlights

- After sequential BPV+Peg-IFN, 9–15% of participants achieved the primary outcome.
- All primary outcome responders had baseline HBsAg ≤3,000 IU/ml.
- BPV+Peg-IFN appeared to reduce relapse rates vs. BPV alone (cross-trial comparison).
- Sequential BPV+Peg-IFN did not convert partial responders to complete responders.
- BPV did not appear to adversely influence the safety profile of subsequent Peg-IFN.

Impact and implications

This phase IIb study investigated whether sequential therapy with bepirovirsen followed by Peg-IFN could improve off-treatment response rates to bepirovirsen alone by converting partial bepirovirsen responders to full responders and/or reducing relapse rates in participants with chronic HBV. These data show that sequential therapy with bepirovirsen followed by Peg-IFN is tolerable and effective; in patients with a bepirovirsen response, sequential treatment with Peg-IFN may help to reduce off-treatment relapses in participants on stable NA. Participants had a similar response during bepirovirsen treatment as seen in B-Clear, with increased response rates in participants with lower baseline HBsAg; all responders to the sequential regimen had baseline HBsAg <3000 IU/mL. As the first study of antisense oligonucleotide-mediated RNA silencing followed by interferon immunomodulation in patients with chronic HBV infection, this study is an important proof-of-concept for sequential therapy, shedding light on the therapeutic potential of utilising immunomodulators following suppression of HBV antigens.

Sequential Peg-IFN after bepirovirsen may reduce post-treatment relapse in chronic hepatitis B

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Background & Aims: Bepirovirsen, an antisense oligonucleotide, induces sustained reductions in hepatitis B surface antigen (HBsAg) and HBV DNA to below the lower limit of quantification (<LLOQ) in a subset of patients. The B-Together study investigated if sequential bepirovirsen and pegylated interferon- α -2a (Peg-IFN) therapy can reduce relapse rates and improve response rates.

Methods: In this phase IIb, multicentre, open-label trial, participants on stable nucleos(t)ide analogue (NA) therapy were randomised 1:1 to bepirovirsen 300 mg once weekly (plus loading dose on Days 4 and 11) for 24 (Arm 1) or 12 (Arm 2) weeks followed by Peg-IFN 180 μ g once weekly for up to 24 weeks, with up to 36 weeks follow-up. Participants continued NA therapy throughout. The primary outcome was the proportion of participants with HBsAg <0.05 IU/ml and HBV DNA <LLOQ for 24 weeks after planned end of Peg-IFN treatment, in the absence of newly initiated antiviral therapy.

Results: The intent-to-treat population included 108 participants (Arm 1, n = 55; Arm 2, n = 53). The primary outcome was achieved by 5 (9%) participants in Arm 1 and 8 (15%) in Arm 2. All responders had baseline HBsAg $\leq$ 3,000 IU/ml. Indirect comparison with the phase IIb study B-Clear indicates that sequential addition of Peg-IFN may reduce the relapse rates previously observed with bepirovirsen alone. The proportions of participants with adverse events and treatment-related adverse events in both treatment windows were similar between treatment arms.

Conclusions: Sequential therapy with bepirovirsen followed by Peg-IFN is tolerable and effective in participants with chronic HBV infection on stable NA therapy. This proof-of-concept trial demonstrates a potential strategy to extend responses to bepirovirsen by reducing relapse.

Clinical Trial Number: NCT04676724.

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Introduction

Chronic HBV infection is a major global health problem, with approximately 820,000 related deaths worldwide in 2019, mostly attributed to cirrhosis and hepatocellular carcinoma.¹ Prognosis may be improved by achieving functional cure,² defined as sustained loss of hepatitis B surface antigen (HBsAg) and unquantifiable HBV DNA 24 weeks following treatment cessation.^{3–5} Current treatment options include long-term nucleos(t)ide analogue (NA) therapy or finite treatment with pegylated interferon- α -2a (Peg-IFN).^{4,6} Functional cure is uncommon (NA only: <3%, Peg-IFN only: <7%) with current treatments.^{4,5,7,8}

Bepirovirsen is an antisense oligonucleotide targeting all HBV RNAs, including pregenomic RNA. Bepirovirsen impacts HBV infection by reducing viral proteins, reducing HBV DNA, and stimulating the immune system, possibly via toll-like receptor activation.^{9–11} In the phase IIb B-Clear trial, bepirovirsen treatment alone for 12 or 24 weeks induced HBsAg and HBV DNA loss sustained for 24 weeks after end of treatment (EoT) in 3–9% of participants with chronic HBV infection receiving stable NA therapy. Response rates at EoT were higher across treatment arms (up to 26%), but up to 75% of participants relapsed after treatment discontinuation. Participants with lower baseline HBsAg levels were more likely to achieve the

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primary endpoint (up to 16% of participants on stable NAs with baseline HBsAg $\leq 1,000$ IU/ml achieved the primary endpoint).¹²

The B-Together study was designed as a proof-of-concept study based on the hypothesis that Peg-IFN sequential therapy could increase response rates by converting bepirovirsen partial responders (those who had a reduction in HBsAg levels but were HBsAg > lower limit of quantification [LLOQ; 0.05 IU/ml] at EoT) to full responders on Peg-IFN treatment and/or by preventing relapse. There is some evidence to suggest that Peg-IFN is more effective in participants with low baseline HBsAg;¹³ if participants had low HBsAg levels as a result of bepirovirsen treatment, Peg-IFN could theoretically be more effective in these participants. The study examined whether sequential treatment with bepirovirsen for 12 or 24 weeks followed by 24 weeks of Peg-IFN could reduce relapse rates and improve responses observed in the B-Clear trial 24 weeks after EoT.

Materials and methods

Participants

B-Together enrolled adult participants with confirmed chronic HBV infection (were HBsAg positive [+]) for ≥ 6 months before screening and currently receiving stable NA therapy (defined as no changes to their NA regimen from at least 6 months prior to screening and with no planned changes to the stable regimen over the duration of the study) except telbivudine. Key inclusion criteria were: eligible for Peg-IFN treatment with plasma or serum HBsAg >100 IU/ml (measured using Roche Elecsys HBsAg II quant II), HBV DNA <90 IU/ml, and alanine aminotransferase (ALT) $\leq 2 \times$ the upper limit of normal (ULN). Key exclusion criteria were: HCV, HDV, or HIV co-infection; cirrhosis; hepatocellular carcinoma; vasculitis.

Peg-IFN eligibility was assessed at screening and bepirovirsen EoT.

Inclusion/exclusion and Peg-IFN eligibility criteria are listed in the supplementary methods.

Trial design

B-Together was a phase IIb, multicentre, randomised, open-label trial to assess the efficacy and safety of sequential treatment with bepirovirsen followed by Peg-IFN in participants with

chronic HBV infection. The trial was conducted from 28 January 2021–17 February 2023 at 52 sites in 11 countries (Table S1).

Participants were randomised 1:1 to treatment Arm 1 or 2 (Fig. 1). Treatment arms were stratified by HBsAg level (HBsAg $\leq 1,000$ / $>1,000$ IU/ml) and hepatitis B e antigen (HBeAg) status (positive [+]/negative [-]). Investigators and participants were unblinded to study intervention throughout the trial. To protect against foreknowledge of pre-randomisation, allocation was concealed by central blocking.

Participants received weekly bepirovirsen 300 mg (plus a loading dose on Days 4 and 11) for 24 (Arm 1) or 12 (Arm 2) weeks (bepirovirsen treatment window) followed by weekly Peg-IFN 180 μ g for up to 24 weeks (Peg-IFN treatment window), if eligible. Participants were followed for an additional 24 (Arm 1) or 36 (Arm 2) weeks (off-treatment) following planned end of Peg-IFN treatment. Participants continued NA treatment throughout the trial.

Eligible participants began Peg-IFN treatment 1 week after the last dose of bepirovirsen; those not immediately able to start Peg-IFN were re-assessed on Weeks 2, 3, 4, 6, 8, 10, and 13 of the Peg-IFN eligibility window. Participants who failed to meet Peg-IFN eligibility criteria did not start Peg-IFN treatment.

Trial oversight

The sponsor, GSK, was involved in the design, conduct, and analysis of the trial. An Independent Data Monitoring Committee reviewed ongoing unblinded safety data. The trial was conducted in accordance with the Declaration of Helsinki, Council for International Organizations of Medical Sciences International Ethical Guidelines for Biomedical Research Involving Human Subjects, Good Clinical Practice guidelines of the International Council for Harmonisation, and all applicable laws and regulations in participating countries. Protocol and amendments were reviewed and approved by local institutional review boards or independent ethics committees. Written informed consent was obtained from all participants. Details on mitigating bias are provided in the [supplementary methods](#).

Outcomes

The primary efficacy outcome was the percentage of participants in each treatment arm who achieved HBsAg and HBV DNA levels

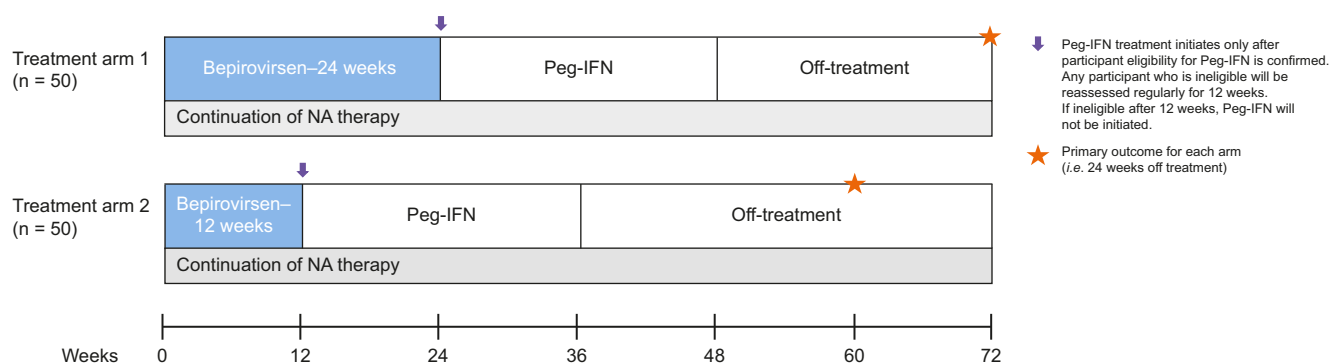


Fig. 1. Trial design. Approximately 50 participants were planned to be enrolled in each treatment arm. The enrolled population included 55 participants in arm 1 and 53 participants in arm 2. Within each arm all participants are expected to end Peg-IFN treatment at the same time, regardless of when they start therapy (Treatment arm 1: all participants should complete treatment by Week 48; Treatment arm 2: Week 36). As such, the planned treatment duration with Peg-IFN for participants may vary from a minimum of 12 weeks to a maximum of 24 weeks. BPV, bepirovirsen; NA, nucleos(t)ide analogue; Peg-IFN, pegylated interferon- α -2a.

below the LLOQ (HBsAg: 0.05 IU/ml, HBV DNA: 20 IU/ml) at every visit for 24 weeks after the planned end of sequential treatment, in the absence of newly initiated antiviral rescue therapy. *Post hoc* analysis evaluated an alternative primary outcome definition assessing the proportion of participants in each arm with HBsAg and HBV DNA <LLOQ at 24 weeks post-EoT. Relapse was defined as any value of HBsAg or HBV DNA $\geq$ LLOQ or newly initiated antiviral rescue therapy.

Secondary objectives included efficacy of bepirovirsen and Peg-IFN sequential therapy on HBV biomarkers (HBsAg and HBeAg) and virus-specific antibody responses (anti-HBeAg status, anti-HBs), durability of virological response, and between-treatment comparison for the primary outcome. Full details of efficacy objectives and estimands are contained in [Table S2](#). Exploratory objectives included generating serum proteomic profiles of enrolled participants ([supplementary methods](#)).

The safety profile of sequential therapy was assessed through clinical assessment of vital signs, laboratory measurements, and adverse event (AE) monitoring. Safety concerns that required additional monitoring were assessed as AEs of special interest (AESIs). AESIs were pre-defined as a group of MedDRA preferred terms ([Table S3](#)), allowing for a comprehensive review of safety data not limited to a specific preferred term, and included ALT increase, vascular inflammation and complement activation, renal injury, and thrombocytopenia for bepirovirsen, and ALT increase, hematologic events, and psychiatric events for Peg-IFN.

Statistical analysis

The trial planned to enrol approximately 100 participants. Using Bayesian inference, this sample size was selected to provide a 65% posterior probability of the true response rate exceeding a fixed threshold under a range of assumed values for the threshold of interest and a range of assumed true response rates.

Primary analysis was conducted when the last participant completed their Week 72 visit. Due to convergence issues caused by quasi-complete separation in the data, the planned Bayesian hierarchical model for the primary analysis was simplified to a Bayesian logistic regression model with fixed effects for treatment arm and baseline stratification factors.

The statistical approach taken for participants who experienced wide disruptive events (WDEs) is as follows. For participants who experienced WDEs (e.g. COVID-19 infections or pandemic lockdowns) resulting in discontinuation or significant interruption/non-adherence to the investigational product (a gap of ≥ 21 days between treatment doses during the on-treatment period except for the Peg-IFN eligibility window), this model utilised multiple imputation (assuming missing at random) to assess the primary outcome. The approach to WDEs was modified post-protocol finalisation in a statistical analysis plan amendment that was finalised prior to database lock; for participants where WDEs caused significant interruptions/non-adherence, but who then returned to the study and were responders on the primary outcome, multiple imputation was not performed, and data were analysed as collected.

A subgroup logistic regression model was constructed to identify significant potential predictors of primary outcome response. To determine additional variables associated with primary outcome response, stratification factors (HBsAg baseline status [$\leq 1,000$ IU/ml; $> 1,000$ IU/ml], HBeAg baseline status

[negative; positive] and treatment arm) were forced into the model, and the following factors were candidates for inclusion using forward stepwise selection: age (years), race (Asian, Black or African American, or White), sex (Male, Female), fibrosis score, BMI (kg/m²), duration of HBV infection (years), type of NA therapy, ALT (IU/L), urine albumin (g/dl), and estimated glomerular filtration rate (ml/sec/1.73 m²). Firth's correction was applied to accurately assess whether the effect of race should be adjusted for in the model.

Efficacy analyses were conducted on the intent-to-treat (ITT) population, which included all randomised participants, analysed based on treatment assignment. Safety analyses were conducted on all randomised participants who received ≥ 1 dose of study treatment, analysed based on treatment received.

Results

Participants

B-Together enrolled 108 participants (Arm 1, $n = 55$; Arm 2, $n = 53$); all were included in the ITT population. Baseline demographics and disease characteristics were similar between arms ([Table 1](#)).¹² Baseline demographics were also similar to the B-Clear study ([Table S4](#)). Most participants had baseline HBsAg $> 1,000$ IU/ml ($n = 74$ [69%]) and were HBeAg- ($n = 80$ [74%]). Nine participants (Arm 1, $n = 6$; Arm 2, $n = 3$) withdrew from the study, two due to AEs ([Fig. S1](#)). All but two participants (one in each Arm) were eligible to begin Peg-IFN treatment following bepirovirsen treatment. COVID-19 infections and lockdowns resulted in missed treatment visits for nine participants (all in Arm 1). HBV DNA levels were typically <LLOQ for the duration of the study for nearly all participants ([Table S5](#)). Intermittent, transient elevations in HBV DNA $\geq$ LLOQ were observed for participants in both treatment arms.

Primary outcome

The primary outcome was achieved in 5 (9%) and 8 (15%) participants in Arm 1 and Arm 2, respectively. The resulting point estimates (95% credible interval) from the Bayesian model were 12% (0, 46%) for Arm 1 and 15% (0, 56%) for Arm 2 ([Table 2](#)). Seven (13%) participants in Arm 1 and 8 (15%) in Arm 2 achieved the alternative primary outcome ([Table 2](#)). The profiles of the two participants who only achieved the alternate primary outcome but not the primary outcome are presented in [Fig. S2](#). For nine participants in Arm 1, WDEs resulted in discontinuation of, significant interruption of, or non-adherence to the investigational product. Of these participants, two missed bepirovirsen dosing (2–4 missed visits) and seven missed Peg-IFN dosing (2–13 missed visits; [Table S6](#)). Two of these participants were virologic responders on Peg-IFN until the WDE occurred. This resulted in interruptions to Peg-IFN treatment that preceded virologic relapse, and both participants were considered failures against the primary outcome ([Table S6](#)).

Irrespective of bepirovirsen treatment duration, a greater proportion of participants with baseline HBsAg $\leq 1,000$ IU/ml or $\leq 3,000$ IU/ml achieved HBsAg and HBV DNA <LLOQ at the end of each treatment window and at 24 weeks after EoT vs. the overall population ([Fig. 2](#)). All participants who met the primary outcome had baseline HBsAg $\leq 3,000$ IU/ml.

Table 1. Baseline demographics and disease characteristics for the enrolled population.

	Arm 1 (n = 55)	Arm 2 (n = 53)	Overall (N = 108)
Age, mean (SD), years	45.5 (10.7)	45.5 (9.4)	45.5 (10.0)
Sex, n (%)			
Female	11 (20)	20 (38)	31 (29)
Male	44 (80)	33 (62)	77 (71)
Race, n (%)			
Asian	30 (55)	26 (49)	56 (52)
Central/South Asian heritage	0	2 (4)	2 (2)
East Asian heritage	23 (42)	15 (28)	38 (35)
Japanese heritage	6 (11)	7 (13)	13 (12)
Southeast Asian heritage	1 (2)	2 (4)	3 (3)
Black or African American	3 (5)	3 (6)	6 (6)
White	22 (40)	24 (45)	46 (43)
Arabic/North African heritage	2 (4)	0	2 (2)
Caucasian/European heritage	20 (36)	24 (45)	44 (41)
HBV viral genotype,* n (%)			
A	3 (5)	6 (11)	9 (8)
B	2 (4)	1 (2)	3 (3)
C	13 (24)	13 (25)	26 (24)
D	5 (9)	5 (9)	10 (9)
E	3 (5)	2 (4)	5 (5)
H	1 (2)	0	1 (<1)
Undetermined	28 (51)	26 (49)	54 (50)
Route of HBV transmission (self-reported), n (%)			
Vertical (mother/child)	7 (13)	13 (25)	20 (19)
From childhood	5 (9)	1 (2)	6 (6)
Sexual contact	0	3 (6)	3 (3)
Blood transfusion	1 (2)	0	1 (<1)
Unknown	40 (73)	34 (64)	74 (69)
Other	2 (4)	2 (4)	4 (4)
BMI, mean (SD), kg/m ²	23.7 (3.4)	25.0 (4.2)	24.3 (3.9)
HBsAg, mean (SD), IU/ml	5,231.4 (8,939.8)	5,218.5 (8,437.4)	5,225.1 (8,656.2)
HBsAg ≤1,000 IU/ml, n (%)	17 (31)	17 (32)	34 (31)
HBsAg >1,000 IU/ml, n (%)	38 (69)	36 (68)	74 (69)
HBV DNA <LLOQ, n (%)	49 (89)	48 (91)	97 (90)
HBeAg status, n (%)			
Negative	41 (75)	39 (74)	80 (74)
Positive	14 (25)	14 (26)	28 (26)
Current NA therapy†, n (%)			
TAF or TDF	33 (61)	37 (70)	70 (65)
Entecavir	19 (35)	12 (23)	31 (29)
Other‡	2 (4)	4 (8)	6 (6)
Duration of current NA therapy, n (%)			
<3 years	26 (48)	27 (51)	53 (50)
≥3 years	28 (52)	26 (49)	54 (50)
Duration of HBV infection, n (%)			
<5 years	10 (18)	11 (21)	21 (19)
≥5–<10 years	7 (13)	10 (19)	17 (16)
≥10–20 years	21 (38)	22 (42)	43 (40)
≥20 years	17 (31)	10 (19)	27 (25)
ALT, n (%)			
≤ULN	51 (93)	43 (81)	94 (87)
>ULN	4 (7)	10 (19)	14 (13)
METAVIR fibrosis score§			
n	7	8	15
F0, n (%)	0	1 (13)	1 (7)
F1, n (%)	4 (57)	4 (50)	8 (53)
F2, n (%)	3 (43)	3 (38)	6 (40)
F3, n (%)	0	0	0

ALT, alanine aminotransferase; BMI, body mass index; HBeAg, hepatitis B e antigen; HBsAg, hepatitis B surface antigen; HBV, hepatitis B virus; LLOQ, lower limit of quantification; NA, nucleos(t)ide analogue; SD, standard deviation; TAF, tenofovir alafenamide fumarate; TDF, tenofovir disoproxil fumarate; ULN, upper limit of normal.

*Genotypes F, G, I or J were not detected in any participant.

†Data on current NA therapy was missing for one participant in Arm 1.

‡Other NA medications included tenofovir, tenofovir disoproxil, lamivudine or tenofovir alafenamide.

§Based on medical history/records.

Table 2. Proportion of participants with primary outcome response 24 weeks after planned end of Peg-IFN treatment.

	Arm 1 (n = 55)	Arm 2 (n = 53)
Overall		
Participants with primary outcome response, n (%)	5 (9)	8 (15)
Bayesian point estimate of primary outcome response rate, [†] % (95% CrI)	12 (0, 46)	15 (0, 56)
By baseline strata		
Stratum 1: HBsAg ≤1,000 IU/ml, HBeAg negative		
n	16	14
Participants with primary outcome response n (%)	3 (19)	7 (50)
Stratum 2: HBsAg >1,000 IU/ml, HBeAg negative		
n	25	25
Participants with primary outcome response n (%)	1 (4)	1 (4)
Stratum 3: HBsAg ≤1,000 IU/ml, HBeAg positive		
n	1	3
Participants with primary outcome response n (%)	1 (100)	0
Stratum 4: HBsAg >1,000 IU/ml, HBeAg positive		
n	13	11
Participants with primary outcome response n (%)	0	0
Post hoc analyses – overall		
Participants achieving alternative primary outcome [†] response, n (%)	7 (13)	8 (15)

Note: with the exception of the Bayesian model, summaries are based on observed data.

CrI, credible interval; HBeAg, hepatitis B e antigen; HBsAg, hepatitis B surface antigen; HBV, hepatitis B virus; LLOQ, lower limit of quantification; Peg-IFN, pegylated interferon-α-2a.

*The point estimate of primary response rate and its 95% highest posterior density CrI are estimated from a Bayesian logistic regression model that incorporates the analysis stratification factors and treatment arm.

[†]The alternative primary outcome was defined as HBsAg and HBV DNA <LLOQ at off-treatment Week 24.

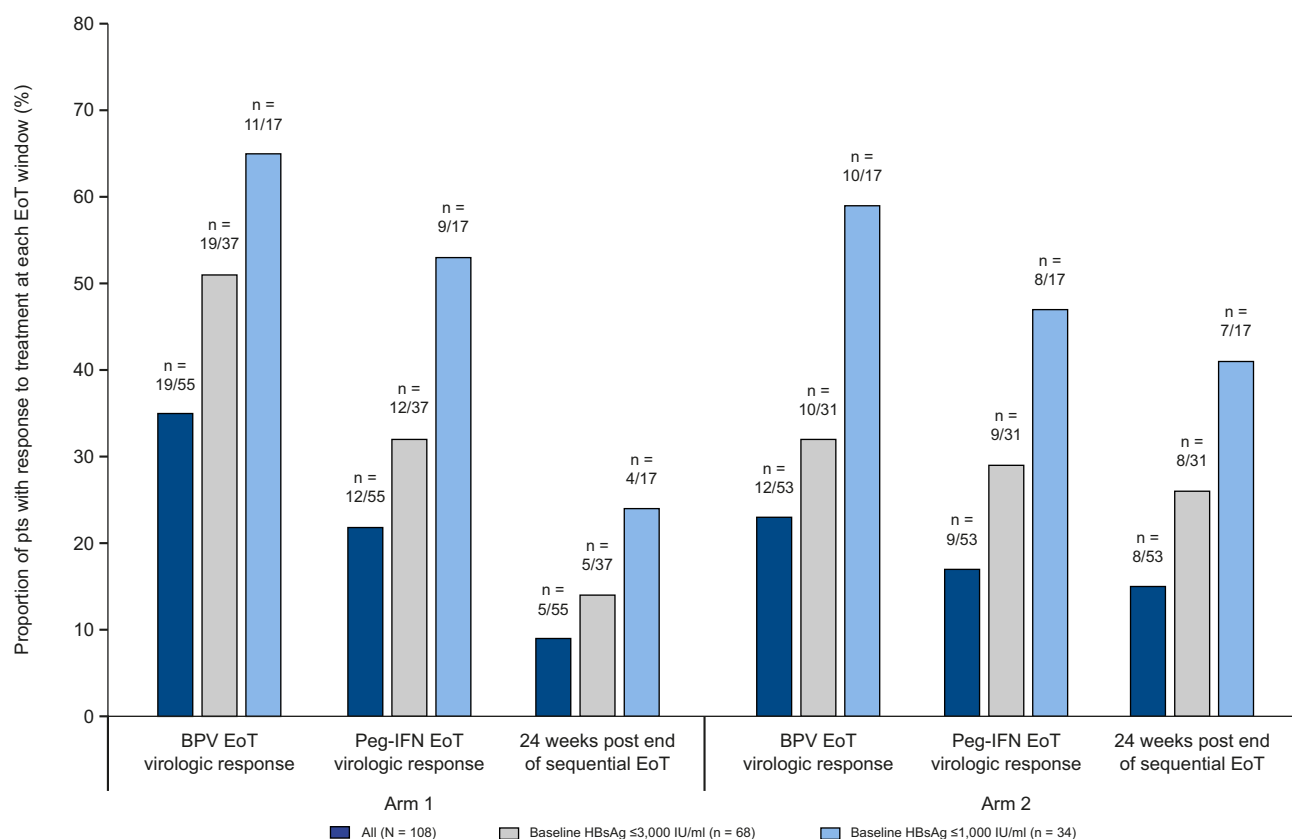


Fig. 2. Proportion of participants with HBsAg and HBV DNA <LLOQ at end of each treatment window and 24 weeks post-EoT: overall, by baseline HBsAg ≤3,000 IU/ml, and by baseline HBsAg ≤1,000 IU/ml. Participants in Arm 1 received weekly bepirovirsen 300 mg per week for 24 weeks (plus a loading dose on Days 4 and 11) followed by Peg-IFN 180 µg per week for up to 24 weeks. Participants in Arm 2 received weekly bepirovirsen 300 mg per week for 12 weeks (plus a loading dose on Days 4 and 11) followed by Peg-IFN 180 µg per week for up to 24 weeks. BPV, bepirovirsen; EoT, end of treatment; HBsAg, hepatitis B surface antigen; HBV, hepatitis B virus; LLOQ, lower limit of quantification; Peg-IFN, pegylated interferon-α-2a; pts, participants.

Table 3 presents the final subgroup logistic regression model consisting of variables selected using a forward selection approach based on minimizing the Akaike information criterion. The final model identified low baseline HBsAg level, high BMI, and treatment with tenofovir alafenamide fumarate or tenofovir disoproxil fumarate as statistically significant predictors of response (Table 3).

Secondary outcomes

HBsAg and HBV DNA levels

At bepirovirsen EoT, 20/55 (36%; Arm 1) and 12/53 (23%; Arm 2) participants achieved HBsAg and HBV DNA <LLOQ. Of these, 8/20 (40%; Arm 1) and 5/12 (42%; Arm 2) relapsed before Peg-IFN EoT. At Peg-IFN EoT, 13/55 (24%; Arm 1) and 9/53 (17%; Arm 2) participants had HBsAg and HBV DNA <LLOQ. Of these, 7/12 (58%; Arm 1) and 0/9 (0%; Arm 2) relapsed during follow-up. Of the nine responders in Arm 2 at Peg-IFN EoT, one failed to achieve the primary outcome due to missing data off-treatment but had no observed off-treatment relapse. A Kaplan-Meier survival plot of time from bepirovirsen EoT to relapse in B-Together and B-Clear is shown in Fig. 3. The proportion of participants within each HBsAg category over time is shown in Fig. 4. Only two participants in Arm 2 who did not achieve HBsAg and HBV DNA <LLOQ at bepirovirsen EoT had HBsAg and HBV DNA loss following Peg-IFN treatment. Both these participants had sharp decreases in HBsAg on bepirovirsen treatment and were just above the LLOQ (0.05 IU/ml and 0.07 IU/ml) at bepirovirsen EoT. Both these participants were likely on their way to HBsAg and HBV DNA <LLOQ irrespective of Peg-IFN (Fig. S3). A flow diagram detailing the proportion of participants with HBsAg or HBV DNA <LLOQ, and those with relapse at each EoT timepoint is available

in the supplementary materials (Fig. S4). Mean change from baseline in HBsAg, HBV DNA, HBeAg, and anti-HBs is shown in Fig. S5. Individual changes in HBsAg over time by baseline HBsAg levels are shown in Fig. S6.

HBeAg and HBe antibody (anti-HBe)

The proportion of participants who converted from HBeAg+ to HBeAg- on-treatment is shown in Fig. S7. Of the participants who were HBeAg+ at baseline, 8/14 (57%) and 1/14 (7%) became anti-HBe+ by the end of the bepirovirsen treatment period in Arm 1 and Arm 2, respectively. At the end of the Peg-IFN treatment period, this changed to 5/9 (56%) and 1/9 (11%), respectively. Only a subset of participants who converted from HBeAg+ to HBeAg- during the study tested positive for anti-HBe (8/11 [73%] across both treatment arms). Most participants maintained their baseline anti-HBe status throughout the study. Some participants seroconverted during the study, while some participants experienced transient reverse seroconversion back to positive during treatment windows. HBeAg status was stable off-treatment to Week 24. One baseline HBeAg+ participant achieved the primary outcome but never developed anti-HBe+. In Arm 2, two participants reverted to HBeAg+ status after Week 24.

Anti-HBs response

Most participants had no detectable anti-HBs at baseline. Among participants who achieved HBsAg and HBV DNA <LLOQ by Peg-IFN EoT, anti-HBs status did not correlate with primary outcome achievement; anti-HBs values were highly variable and overlapping in those who relapsed and those who maintained HBsAg <LLOQ (Fig. S8).

Table 3. Logistic regression of variables potentially associated with response.

Parameter	Participants, n/N (%)	Odds ratio (95% CI)	p values
Arms			
Arm 1 (reference)	5/54 (9.3)	1.01 (0.17, 5.96)	0.99
Arm 2	8/53 (15.1)		
Baseline HBsAg			
≤1,000 IU/ml (reference)	11/34 (32.4)	0.05 (0.00, 0.26)	<0.0001
>1,000 IU/ml	2/73 (2.7)		
Baseline HBeAg status			
Negative (reference)	12/79 (15.2)	0.28 (0.01, 3.27)	0.3263
Positive	1/28 (3.6)		
Age	Not applicable/107	1.05 (0.97, 1.15)	0.2580
Race			
White (reference)	10/46 (21.7)		
Asian	3/56 (5.4)	0.43 (0.06, 3.14)	0.3884
Black or African American	0/5 (0)	0.01 (<0.01, 1.09)	0.0539
BMI (kg/m ²)	Not applicable/107	1.34 (1.05, 1.93)	0.0183
Duration of HBV infection	Not applicable/107	1.03 (0.97, 1.10)	0.2928
Type of NA			
TAF & TDF (reference)	10/70 (14.3)		
Entecavir	2/31 (6.5)	0.09 (0.01, 0.61)	0.0119
Other	1/6 (16.7)	0.95 (0.06, 11.66)	0.9703
ALT (IU/ml)	Not applicable/107	0.95 (0.87, 1.01)	0.0927
Urine albumin (mg/L)	Not applicable/107	0.99 (0.91, 1.09)	0.8125

Odds ratios, 95% CIs and p values calculated using a subgroup logistic regression model adjusting for variables showed in this table. The final model was constructed by selecting from the following variables using a forward stepwise selection approach, based on AIC: age (years), race (Asian; Black or African American; White), sex (male; female), BMI (kg/m²), fibrosis score, duration of HBV infection (years), type of NA, ALT (IU/L), urine albumin (g/dl), eGFR (ml/sec/1.73 m²). Baseline HBsAg (≤3, >3 log₁₀ IU/ml), HBeAg status (positive, negative) and Arm are included in the model regardless of their impact on AIC.

AIC, Akaike information criterion; ALT, alanine aminotransferase; BMI, body mass index; eGFR, estimated glomerular filtration rate; HBeAg, hepatitis B e antigen; HBsAg, hepatitis B surface antigen; HBV, hepatitis B virus; n, number of participants with non-missing baseline value (after model selection); NA, nucleos(t)ide analogue; TAF, tenofovir alafenamide fumarate; TDF, tenofovir disoproxil fumarate.

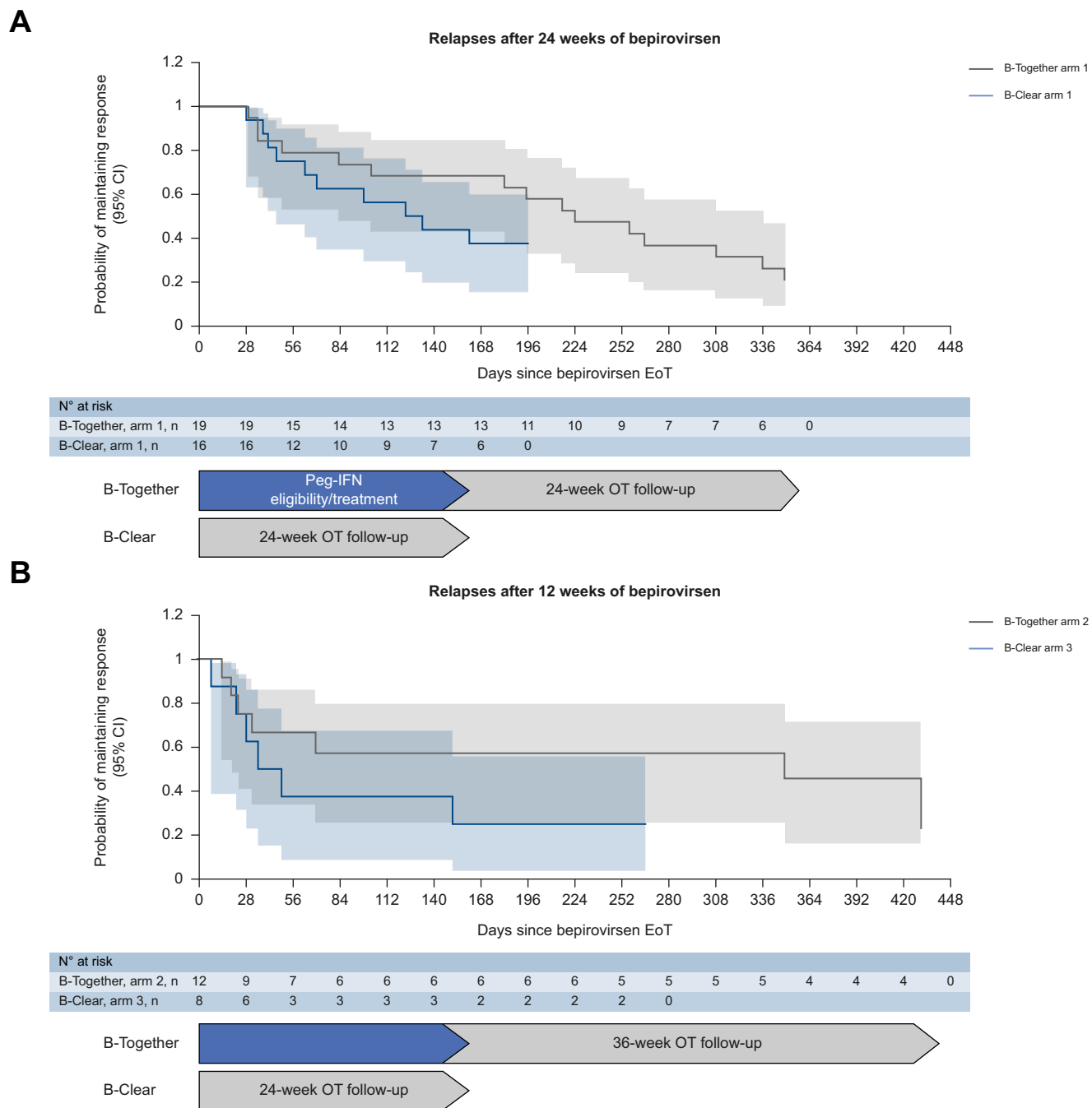


Fig. 3. Kaplan-Meier plot of time from end of bepirovirsen treatment to relapse in B-Together and B-Clear after 24 weeks and 12 weeks of bepirovirsen. Relapse is defined as any value of HBsAg or HBV DNA >LLOQ or newly initiated antiviral rescue therapy. One participant in Arm 1 had HBsAg and HBV DNA <LLOQ at bepirovirsen EoT but met study definition of rescue medication and is not included in this analysis. Two participants in Arm 2 were primary outcome responders but did not have HBsAg and HBV DNA <LLOQ at bepirovirsen EoT and were not included in this analysis. Solid lines represent Kaplan-Meier estimates; shaded areas represent 95% CIs. CI, confidence interval; EoT, end of treatment; HBsAg, hepatitis B surface antigen; HBV, hepatitis B virus; LLOQ, lower limit of quantification; OT, off-treatment; Peg-IFN, pegylated interferon- α -2a.

ALT levels

Transient increases in ALT levels $\geq 3 \times$ ULN were seen in 38% of participants in Arm 1 and 40% in Arm 2; 24% and 19% of participants, respectively, experienced such increases during the bepirovirsen treatment window, and 18% and 25%, respectively, during the Peg-IFN treatment window. During the

bepirovirsen treatment window, ALT increases coincided with a decline in HBsAg, whereas there was no association between HBsAg and ALT levels during the Peg-IFN treatment window (Fig. S9). By off-treatment Week 2, most participants (Arm 1: 86%, Arm 2: 84%) had ALT <3x ULN. By off-treatment Week 12, all participants had ALT <2x ULN, with 82% $\leq$ ULN.

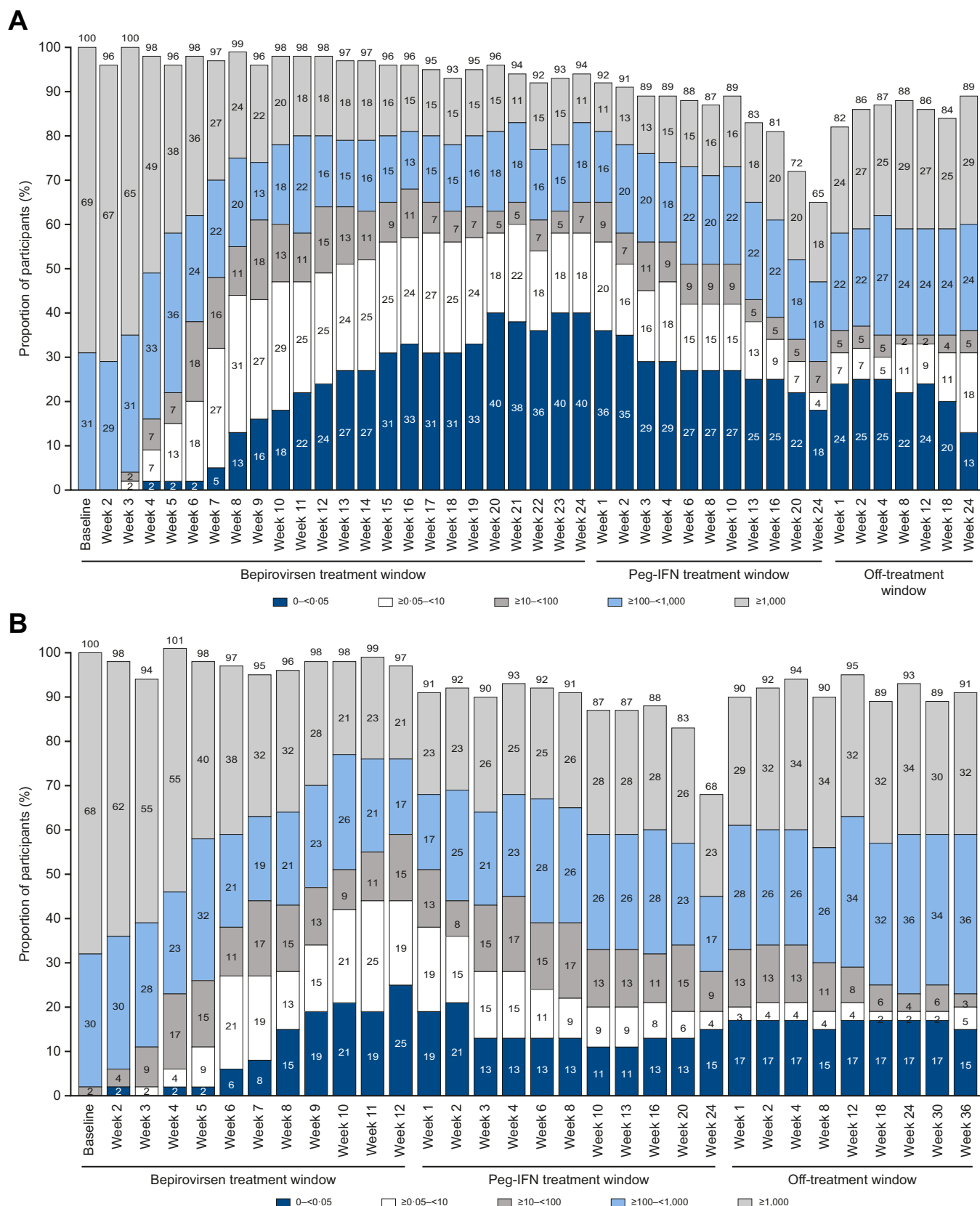


Fig. 4. Proportion of participants with HBsAg <0.05, ≥0.05–<10, ≥10–<100, ≥100–<1,000, or ≥1,000 IU/ml in (A) Arm 1 and (B) Arm 2. HBsAg, hepatitis B surface antigen; Peg-IFN, pegylated interferon- α -2a.

Table 4. Adverse events during each treatment window.

	Arm 1 (n = 55)	Arm 2 (n = 53)	Total (N = 108)
On-treatment bepirovirsen			
Any AE, n (%)	48 (87)	42 (79)	90 (83)
Any AE related to study treatment	47 (85)	39 (74)	86 (80)
AEs leading to permanent discontinuation	1 (2)	2 (4)	3 (3)
Any grade 3 and 4 event*	20 (36)	16 (30)	36 (33)
ALT increase	6 (30)	4 (25)	10 (9)
Creatinine renal clearance decrease	0	3 (19)	3 (3)
Injection-site erythema	0	3 (19)	3 (3)
Any SAE, n (%)†	4 (7)	1 (2)	5 (5)
Any SAE related to study treatment	1 (2)	0	1 (<1)
AESIs for bepirovirsen, n (%)‡			
ALT increase	16 (29)	8 (15)	24 (22)
Vascular inflammation and complement activation	16 (29)	6 (11)	22 (20)
Thrombocytopenia	11 (20)	7 (13)	18 (17)
Renal injury	10 (18)	6 (11)	16 (15)
Additional AEs monitored for Peg-IFN, n (%)			
Neutropenia	1 (2)	3 (6)	4 (4)
Leukopenia	1 (2)	1 (2)	2 (2)
Peg-IFN eligibility window			
Any AE, n (%)	6 (11)	6 (11)	12 (11)
Any AE related to study treatment	3 (5)	3 (6)	6 (6)
AEs leading to permanent discontinuation	0	0	0
Any grade 3 and 4 event*	2 (4)	0	2 (2)
Any SAE, n (%)†	0	0	0
Any SAE related to study treatment	0	0	0
AESIs for bepirovirsen, n (%)‡			
ALT increase	0	1 (2)	1 (<1)
Vascular inflammation and complement activation	1 (2)	0	1 (<1)
Thrombocytopenia	1 (2)	0	1 (<1)
Renal injury	1 (2)	3 (6)	4 (4)
Additional AEs monitored for Peg-IFN, n (%)			
Neutropenia	1 (2)	2 (4)	3 (3)
Leukopenia	1 (2)	0	1 (<1)
On-treatment Peg-IFN			
Any AE, n (%)	46 (84)	43 (81)	89 (82)
Any AE related to study treatment	46 (84)	42 (79)	88 (81)
AEs leading to permanent discontinuation	1 (2)	0	1 (<1)
Any grade 3 and 4 event*	17 (31)	8 (15)	25 (23)
Neutrophil count decrease	5 (29)	2	7 (6)
Neutropenia	4	2	6 (6)
Aspartate aminotransferase increase	2	1	3 (3)
Any SAE, n (%)†	0	0	0
Any SAE related to study treatment	0	0	0
AESIs for bepirovirsen, n (%)‡			
ALT increase	16 (29)	5 (9)	21 (19)
Vascular inflammation and complement activation	6 (11)	4 (8)	10 (9)
Thrombocytopenia	21 (38)	16 (30)	37 (34)
Renal injury	3 (5)	3 (6)	6 (6)
Additional AEs monitored for Peg-IFN, n (%)			
Neutropenia	10 (18)	11 (21)	21 (19)
Leukopenia	10 (18)	6 (11)	16 (15)
Post treatment			
Any AE, n (%)	31 (56)	31 (58)	62 (57)
Any AE related to study treatment	9 (16)	9 (17)	18 (17)
AEs leading to permanent discontinuation	0	0	0
Any grade 3 and 4 event*	3 (5)	0	3 (3)
Any SAE, n (%)†	2 (4)	1 (2)	3 (3)
Any SAE related to study treatment	0	0	0
AESIs for bepirovirsen, n (%)‡			
ALT increase	2 (4)	1 (2)	3 (3)
Vascular inflammation and complement activation	1 (2)	3 (6)	4 (4)
Thrombocytopenia	2 (4)	1 (2)	3 (3)
Renal injury	4 (7)	2 (4)	6 (6)
Additional AEs monitored for Peg-IFN, n (%)			
Neutropenia	3 (5)	0	3 (3)
Leukopenia	2 (4)	0	2 (2)

(continued on next page)

Table 4. (continued)

	Arm 1 (n = 55)	Arm 2 (n = 53)	Total (N = 108)
Overall			
Any AE, n (%)	52 (95)	52 (98)	104 (96)
Any AE related to study treatment	52 (95)	48 (91)	100 (93)
AEs leading to permanent discontinuation	2 (4)	2 (4)	4 (4)
Any grade 3 and 4 event*	31 (56)	23 (43)	54 (50)
Any SAE, n (%) [†]	6 (11)	2 (4)	8 (7)
Any SAE related to study treatment	1 (2)	0	1 (<1)
AESIs for bepirovirsen, n (%) [‡]			
ALT increase	27 (49)	11 (21)	38 (35)
Vascular inflammation and complement activation	19 (35)	11 (21)	30 (28)
Thrombocytopenia	29 (53)	19 (36)	48 (44)
Renal injury	10 (18)	7 (13)	17 (16)
Additional AEs monitored for Peg-IFN, n (%)			
Neutropenia	10 (18)	12 (23)	22 (20)
Leukopenia	10 (18)	7 (13)	17 (16)

AE, adverse event; AESIs, AEs of special interest; ALT, alanine aminotransferase; Peg-IFN, pegylated interferon- α -2a; SAE, serious adverse event.

*AEs were graded according to the DAIDS corrected version 2.1. Grade 1 indicates a mild event, Grade 2 a moderate event, Grade 3 a severe event, Grade 4 a potentially life-threatening event, and Grade 5 death.

[†]Arm 1: bepirovirsen on-treatment window: COVID-19 (n = 1 [2%]), cholecystitis (n = 1 [2%]), cholelithiasis (n = 1 [2%]), drug-induced liver injury (n = 1 [2%]) — considered related to treatment in the opinion of the investigator, lung adenocarcinoma (n = 1 [2%]); post-treatment: appendicitis (n = 1 [2%]), chest injury (n = 1 [2%]) — Arm 2: bepirovirsen on-treatment window: peritonsillar abscess (n = 1 [2%]); post-treatment: spinal stenosis (n = 1 [2%]).

[‡]AESIs were specific to bepirovirsen but were reported and reviewed throughout the entire study period. Four participants permanently discontinued treatment due to an AE (1 dermatitis allergic and 2 injection-site reactions during bepirovirsen treatment; 1 depression during Peg-IFN treatment).

Durability of primary outcome response in Arm 2

All but one participant in Arm 2 who achieved the primary outcome sustained HBsAg and HBV DNA <LLOQ until off-treatment Week 36.

Comparison of bepirovirsen treatment durations

A high degree of variability in both treatment arms, indicated by wide overlapping confidence intervals (CI), prevented a clear conclusion on the optimal bepirovirsen treatment duration to include in sequential therapy with Peg-IFN (Fig. S10).

Safety

The proportion of participants who experienced AEs in the bepirovirsen and Peg-IFN treatment windows was similar between treatment arms (95% vs. 98%), with most AEs considered treatment related (Table 4). Few AEs led to treatment discontinuation (4%; 3 during bepirovirsen and 1 during Peg-IFN treatment). The safety profile of Peg-IFN was aligned with its approved product information. Most AEs reported were captured as AESIs (Table 4 and supplementary results). There were no reports of serious liver injury, clinically significant bleeding events, renal injury, or vascular injury linked to alternate complement pathway activation. Nine serious AEs (SAEs) were reported in eight participants, five during bepirovirsen treatment (one considered treatment related) and three in the off-treatment period. No deaths were reported. The most common AEs were injection-site erythema (Table S7). Details of the five participants who required rescue medication are reported in Table S8.

Grade 3 and 4 events occurred in 33% of participants during the bepirovirsen period and 23% in the Peg-IFN period. Some Grade 3/4 AEs were driven by Division of AIDS Table for Grading the Severity of Adult and Pediatric Adverse Events (DAIDS) grading for laboratory changes and may not have been correlated with clinical severity. ALT increases of 5–10x ULN were considered Grade 3 and >10x ULN was considered Grade 4, according to DAIDS. There were no life-threatening events in the study. For AESIs associated with increased ALT, a larger

proportion of participants reported events in Arm 1 compared with Arm 2 (49% vs. 21%). The majority of events were reported as preferred terms of ALT increased and aspartate aminotransferase increased. In each treatment arm, the proportion of participants reporting events was comparable between the bepirovirsen and Peg-IFN periods (Arm 1: 29% vs. 29% and Arm 2: 15% vs. 9%; Table 4). Increases in ALT levels observed throughout the study were asymptomatic and resolved without increases in bilirubin. One ALT increase was reported as an SAE of drug-induced liver injury. The participant was not hospitalised for the event and reported no signs or symptoms associated with the ALT increase. The treatment was held at Week 7 and withdrawn at Week 9, when the participant met the stopping criteria per protocol (ALT >10x ULN for ≥ 4 weeks). An ultrasound of the liver showed no changes compared with a scan performed prior to start of the study. Liver event work-up did not reveal any alternative causes of ALT elevations. During bepirovirsen treatment, the increase in ALT occurred within the first 12 weeks, mainly from Week 5 onwards (Fig. S11A). Also notable is that there was large variability around the increase in ALT during this time period. There does not appear to be an association between maximum ALT increase and maximum decline in HBsAg (Fig. S11B,C). Other AESIs are discussed in the supplementary results.

Discussion

In B-Together, bepirovirsen 300 mg/week for 12–24 weeks followed by up to 24 weeks of Peg-IFN 180 μ g/week resulted in 9–15% of participants achieving the primary outcome. Consistent with findings from B-Clear,¹² participants with lower baseline HBsAg were more likely to respond to bepirovirsen treatment. All primary outcome responders had baseline HBsAg <3,000 IU/ml, and the proportions of participants with HBsAg and HBV DNA <LLOQ were highest in participants with baseline HBsAg $\leq 1,000$ IU/ml. At the time of B-Together study design, it was unknown what participant HBsAg level would be associated with the most benefit from sequential bepirovirsen and Peg-IFN treatment. The reduced efficacy in participants

with higher baseline HBsAg levels is an important observation of our study, which will prevent unnecessary dosing of Peg-IFN in participants with high baseline HBsAg. These data are consistent with data from the Everest study, where participants with low baseline HBsAg (<500 IU/ml) were more likely to respond to Peg-IFN treatment. Comparisons of the Everest study with B-Together are difficult as Everest enrolled participants with much lower baseline HBsAg levels (<1,500 IU/ml) than those enrolled in B-Together.¹⁴ Furthermore, data from model-based meta-analyses also indicate that lower baseline HBsAg enhances responses to Peg-IFN treatment.¹³

In contrast with B-Clear study results,¹² the proportion of participants who achieved the primary outcome was numerically higher with 12-week (Arm 2) vs. 24-week bepirovirsen treatment (Arm 1). The observed difference in efficacy between arms appeared to be driven by a higher relapse rate in Arm 1 during the Peg-IFN treatment window and during off-treatment follow-up. However, it is difficult to fully compare the differences in arms, owing to both the relatively small study size and the heterogeneity observed in the study population. Due to the small participant numbers in each arm and a high level of variability (wide and overlapping 95% CIs), there was uncertainty regarding between-treatment differences. Additionally, Arm 1 was disproportionately impacted by COVID-19 lockdowns during the Peg-IFN eligibility and treatment periods. Arm 2 was not impacted during on-treatment weeks as the Peg-IFN treatment window had ended. Finally, additional aspects such as HBV genotype, route of transmission, duration of infection and age at infection could have impacted the efficacy of Peg-IFN, but the small study size makes it difficult to fully assess determinants of response.^{4,15–17}

While one study has shown that combination therapy of Peg-IFN and NA for 48 weeks could yield 9% response rates, the duration of Peg-IFN treatment greatly impacted this response; 16 weeks of combination therapy resulted in only a 2.8% response rate.¹⁸ This study suggests that the duration of Peg-IFN treatment impacts responses to treatment. In B-Together, there was little evidence to show that Peg-IFN combined with NAs could further reduce HBsAg levels following bepirovirsen treatment. Of participants who were responders at Peg-IFN EoT, HBsAg loss was achieved in all but two participants during bepirovirsen treatment. Peg-IFN treatment instead seemed to improve the maintenance of HBsAg and HBV DNA loss off-treatment, which appears to be driven by a Peg-IFN-associated prevention of relapse in some bepirovirsen responders. In B-Clear, 63% and 75% participants relapsed off-treatment after 24-week and 12-week bepirovirsen 300 mg treatment, respectively. Here, 58% (Arm 1) and 0% (Arm 2) of participants who achieved HBsAg and HBV DNA <LLOQ at Peg-IFN EoT relapsed off-treatment. More than half (58%) of participants who were responders at bepirovirsen EoT maintained response on Peg-IFN treatment. Two participants in Arm 2 were responders at Peg-IFN EoT but not responders at bepirovirsen EoT. For all participants who were responders at bepirovirsen EoT but failed to achieve response at Peg-IFN EoT, the reason for failure was virological relapse. These results indicate sequential treatment with bepirovirsen and Peg-IFN in participants on NA therapy may improve seroclearance rates seen with NA (<3%) or Peg-IFN (<7%) treatment alone and NA/Peg-IFN combination therapy (9%).^{5,7,8,18}

Sequential therapy did not highlight any new or cumulative safety signals for bepirovirsen. Sequential regimens of either

12- or 24-week bepirovirsen followed by 24-week Peg-IFN showed a similar safety profile, with no new safety concerns. Bepirovirsen did not appear to adversely influence the safety profile of subsequent Peg-IFN. Injection-site erythema, pyrexia, and ALT increases were the most common AEs reported. ALT increases can be seen in patients with chronic HBV infection due to disease activity, immune clearance of infected hepatocytes, or drug-induced liver injury.^{11,19,20} As in the B-Clear and phase IIa studies,¹² increases in ALT during bepirovirsen treatment coincided with decline in HBsAg; during the Peg-IFN treatment window, there was no association between HBsAg and ALT levels. There was one SAE related to an ALT increase (see [supplementary materials](#)). No serious liver injury occurred, but the participant met protocol-defined permanent discontinuation criteria. The event was asymptomatic, as were all other events associated with increased ALT. A decline in platelet count was observed over the course of sequential treatment that improved in the follow-up period; compared with Peg-IFN, the influence of bepirovirsen was minimal.

Some limitations should be noted. Firstly, the B-Together study design lacks control groups, *i.e.* bepirovirsen 300 mg for 12 or 24 weeks alone, which limits comparisons of bepirovirsen and Peg-IFN sequential therapy with bepirovirsen alone. Thus, the study uses indirect comparisons to the B-Clear study. While indirect comparisons have their own inherent limitations, owing to potential differences in baseline characteristics and population heterogeneity between studies, the study team mitigated the impact of this indirect comparison by recruiting from the same sites (73.5% of the sites from B-Clear are the same as those in B-Together), and both studies had the same inclusion and exclusion criteria, with the exception of the Peg-IFN eligibility criteria. The inherent heterogeneity of patients with HBV impacts comparisons across and within studies unless very large study numbers are available. Baseline characteristics were similar across both studies ([Table S4](#)), and all data have been entered into a pharmacokinetic/pharmacodynamic model that compares B-Together with B-Clear; the model (unpublished data) shows that on-treatment responses during the bepirovirsen treatment window in B-Together are in line with B-Clear.

Secondly, the COVID-19 pandemic disproportionately affected Arm 1 during the Peg-IFN treatment period, with nine participants missing 50 Peg-IFN dosing visits on Arm 1. Despite randomisation and a similar start date, Arm 1 was impacted owing to the longer bepirovirsen treatment window. Arm 2 participants had already completed their dosing visits when lockdowns began. This complicated the comparison of Arm 1 and 2. The Bayesian logistic regression model utilised multiple imputation to mitigate the effect of confounding on the primary outcome caused by missed dosing. However, the modelled point estimate may still have underestimated the response rate had this confounding not occurred. Secondly, the primary outcome definition of HBsAg and HBV DNA <LLOQ maintained at all timepoints off-treatment is highly conservative and likely underestimated the proportion of participants with response. Therefore, the less conservative and simpler alternative primary outcome of HBsAg and HBV DNA <LLOQ at off-treatment Week 24 was applied.

Taken together, these data show sequential therapy with bepirovirsen followed by Peg-IFN is tolerable and effective in participants with chronic HBV infection on stable NA therapy.

Participants had a similar response to bepirovirsen treatment as seen in B-Clear,¹² with increased response rates in participants with lower baseline HBsAg.¹² Peg-IFN did not convert partial responders to full responders, but may have had some impact on reducing relapses. While the study did not show the substantive improvement in bepirovirsen response we hypothesised, further studies of sequential or combination therapy with bepirovirsen and other agents such as HBsAg-lowering agents (e.g. siRNA) or different immunomodulators are being considered to improve

response rates and increase the treatable population, and ultimately deliver functional cure to more patients.

Overall, this trial highlights sequential therapy as a valid treatment strategy for the finite treatment of HBV infection.^{21,22} Long-term durability of response will be evaluated in the B-Sure trial (NCT04954859),²³ which will follow responders from B-Together and B-Clear for an additional 33 months and allows for cessation of NA therapy and assessment of functional cure.

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Abbreviations

AE, adverse event; AESIs, AEs of special interest; AIC, Akaike information criteria; ALT, alanine aminotransferase; CI, confidence intervals; DAIDS, Division of AIDS table for grading the severity of adult and pediatric adverse events; EoT, end of treatment; FC, functional cure; HBeAg, hepatitis B e antigen; HBsAg, hepatitis B surface antigen; ITT, intent-to-treat; LLOQ, lower limit of quantification; MedDRA, Medical dictionary for regulatory activities; NA, nucleos(t)ide analogue; Peg-IFN, pegylated interferon- α -2a; PT, preferred terms; SAE, serious AE; ULN, upper limit of normal; WDEs, wide disruptive events.

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Conflict of interest

Roche Elecsys is a registered trademark of F. Hoffmann-La Roche Ltd.

MB has received grant support from Gilead and AbbVie, consultant fees from Gilead, AbbVie, Roche, Altimmune, and GSK, and lecture fees from Altimmune. JH has received grant support from Gilead and Roche, has received consultant fees from Roche, and lecture fees from AbbVie Korea, Roche, Yuhan Korea, Oncolys and Gilead, and is a member of an AstraZeneca steering committee. YT received grant funding from AbbVie, FUJIREBIO Inc., Sysmex Corp, Janssen Pharmaceutical, GSK, and Gilead, and speaking honoraria from AbbVie, Gilead, Chugai Pharmaceutical, ASKA Pharmaceutical, OTSUKA Pharmaceutical, Takeda Pharmaceutical Co., and GSK. PA declares no conflicts of interest. MA declares no conflicts of interest. JC has received grant support from Gilead and AbbVie, consultant fees from Gilead, and lecture fees from Gilead and AbbVie. EC has received grant support from GSK and Target RWE. CSC has received grant support from Gilead, Janssen, and Altimmune, consultant fees from Gilead, Roche, and Altimmune (paid to the University of Calgary), and lecture fees from Gilead. KF declares no conflicts of interest. NG declares no conflicts of interest. SCG has received grant support from AbbVie Pharmaceuticals, Arbutus, Cyma-Bay, DURECT, Genfit, Gilead Sciences, GSK, Hightide, Intercept, Mirum, Pliant, and Merck and consultant fees from Gilead, AbbVie, and CymaBay. EJ has received grant support from AbbVie, Bristol Myers Squibb, Gilead, Immunocore,

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Please refer to the accompanying ICMJE disclosure forms for further details.

Authors' contributions

The authors meet criteria for authorship as recommended by the International Committee of Medical Journal Editors, take responsibility for the integrity of the work as a whole, contributed to the writing and reviewing of the manuscript, and have given final approval for the version to be published. Leigh Felton, Maximilian Lee, Thomas Greene, Grant McGonagle, Helene Plein, Amir Youssef, Dickens Theodore, Melanie Paff, Rob Elston, Jessica Lim, and Stuart Kendrick, contributed to the conception and design of the study. Divya Lakshminarayanan, Leigh Felton, Maximilian Lee, Thomas Greene, Helene Plein, Amir Youssef, and Jessica Lim contributed to data analysis. Maria Buti, Pietro Andreone, Masanori Atsukawa, Eric Chak, Carla S. Coffin, Kei Fujiwara, Natalya Gankina, Stuart C. Gordon, Jeong Heo, Ewa Janczewska, Atsumasa Komori, Pietro Lampertico, Stuart McPherson, Vyacheslav Morozov, Robert Plesniak, Sébastien Poulin, Ryan Pablo, Olga Sagalova, Guoping Sheng, Yasuhito Tanaka, Natalya Voloshina, Qing Xie, Joaquín Cabezas, Hyung Joon Yim, and Divya Lakshminarayanan contributed to the acquisition of data. Maria Buti, Jeong Heo, Joaquín Cabezas, Hyung Joon Yim, Divya Lakshminarayanan, Leigh Felton, Susan Dixon, Maximilian Lee, Thomas Greene, Grant McGonagle, Helene Plein, Amir Youssef, Dickens Theodore, Melanie Paff, Rob Elston, Jessica Lim, and Stuart Kendrick contributed to data interpretation.

Data availability statement

Please refer to GSK weblink to access GSK's data sharing policies and as applicable seek anonymised subject level data via the link <https://www.gsk-studyregister.com/en/>.

Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhep.2024.08.010>.

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Author names in bold designate shared co-authorship

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